

Harshitha Vutukuru Muralikrishna

harshitharaghu22@gmail.com

+91 8328543948

[Portfolio](#)

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[LinkedIn](#)

PROFESSIONAL SUMMARY

A **data science** and **machine learning** professional with more than three years of experience in **building machine learning models** and **ETL pipelines** for the cloud based deployments. Experienced in Python, PyTorch, Sci-kit learn, and AWS. Have excelled in providing **data-driven solutions** that improve accuracy and reduce latency.

EDUCATION

California State University, Los Angeles

Jan 2023 - Dec 2024

Master of Science in Computer Science

Relevant Coursework : Data Science, Machine Learning, Advanced Algorithms, Advanced Artificial Intelligence,

Advanced Web Programming, Operating Systems, Computer Networks

RNS Institute of Technology, Bengaluru

Jul 2015 - Jun 2019

Bachelor of Engineering in Information Science and Engineering

Relevant Coursework : Algorithms, Operating Systems, Data Science, Computer Networks, Artificial Intelligence

TECHNICAL SKILLS

- Programming: Python, SQL, JavaScript, NumPy, Pandas
- Machine Learning and Deep Learning: PyTorch, TensorFlow, Scikit-learn, Keras, PySpark
- NLP and LLMs: LangChain, Hugging face Transformers
- Cloud and MLOps : AWS SageMaker, Kubernetes, Docker, Jenkins, CI/CD pipelines
- Databases : MySQL, SQLite
- Frameworks and Tools: Git, RESTful APIs, Spring Boot

PROFESSIONAL EXPERIENCE

Accenture - Application Development Analyst | Bengaluru, India | Sep 2019 - May 2022

- Handled data for downstream modeling tasks by performing **data cleaning and pre-processing** using Pandas, NumPy, SciKit-learn libraries.
- Enhanced model accuracy and reduced false positives for large financial datasets by **developing classification and regression models** including **Linear Regression, Logistic Regression, Random Forests, XGBoost** for fraud detection and anomaly handling.
- Served large scale **feature extraction by building data transformation pipelines** for efficient data training of the model and for smoother production deployments.
- Led **MLOps processes** using **docker and Kubernetes**. This has shown improved scalability and reduced deployment time.
- Constructed automated **model monitoring on AWS Sage Maker** for detecting performance drift and for maintaining accuracy of the deployed models.
- **Applied Hyperparameter tuning, cross-validation**, and feature importance analysis to enhance model generation and interpretability.

Spectrus Informatics - Full Stack Intern | Bengaluru, India | Feb 2018 - Jul 2018

- Improved user transaction workflows for data driven e-commerce platform using **Angular-2, JavaScript, and SQL**.
- Handled **SQL databases and CRUD API's** to improve data transaction's efficiency and also establishing a foundation for **future analytics and machine learning integration**.

PROJECTS

Brain Tumor Detection using CNNs

- An accuracy of 98% was achieved on MRI based brain tumor detection by **fine-tuning deep CNN architectures** like **ResNet50, VGG19 and HALNet variants**.
- Adopted image pre-processing techniques like **augmentation, normalization and edge detection** for improving the model efficiency and reducing false positives.
- Utilized **ROC-AUC, Recall and F1 score** for evaluating the model's best performance and for obtaining higher accuracy.
- Optimized training pipeline for efficient GPU utilization and reproducibility.
- GitHub Link : https://github.com/harshithavutukuru/Datascience_project/tree/main

AWS SageMaker ML Deployment

- **Developed an end-to-end regression pipeline** using Kaggle's House Prices dataset. The preprocessing and **feature engineering** stored in **Amazon S3**.
- Trained the **XGBoost regression** model with **AWS SageMaker**, **optimized the hyperparameters** with Automatic model tuning which improved the accuracy by 15%.
- **Deployed the model as a real-time endpoint** and integrated with **AWS Lambda** to deliver **low-latency price predictions**.
- GitHub Link : <https://github.com/harshithavutukuru/AWS-SageMaker>

CERTIFICATIONS

AWS Certified Machine Learning – Specialty